AI Interaction & Action Log — Savannah with AI

Date: 29 October 2026 | **Tool:** Codex (course workspace) | **Scope:** Factual session summary, not a verbatim transcript. No subagents used.

User request

“Implement the African Savannah with AI”: (1) specify Unit 1 model before AI, (2) build incrementally with working commits, (3) compare line by line with handwritten source, (4) extend with zebra grazing and lion energy, (5) submit a one-page IMRaD report plus AI log. One user prompt initiated the session; feature steps below were agent actions, not separate user prompts.

Starting evidence & limitations

Existing Git history included **6fb5661** (Specification) and **cb3039c** (F1: Lions and zebras). The original handwritten source was **assignement1/savannah (1).gaml**. Earlier AI exchanges and historic test runs were unavailable; the specification commit does not establish that it preceded the first-ever AI prompt. Existing user files were preserved in a separate **verified_submission** workflow.

Incremental implementation and recorded commits

Step	Action / evidence	Commit
Specify	Read prior spec; preserve all-prey kills and uncapped grass; define extension assumptions	00124c8
F2	Explicit 100 m world; 20 lions and 20 zebras move; 20-step run passed	2fd8236
F3	Predation within 2 m; zebras 20 → 17 in 20 steps (seed 42)	f926313
F4	50 × 50 grass cells; mean 5.109 → 7.009 kg, +0.1/step	10f4860
F5	Population and mean-grass charts; dynamics unchanged	fa6f728
E1	Zebra grazing up to 1 kg/cell; biomass nonnegative	128a3d9
E2	Lion starts at 50 energy; −0.05/wander, +10/zebra, death at ≤0	4e7e456
Verify	Disable parallel execution; correct log decoding; rerun acceptance tests	362d738

Comparison, errors and corrections

Compared the actual handwritten model line by line. Corrected older claims that Unit 1 explicitly declared a 100 m square or included a mean-grass chart. Replaced the earlier AI extension’s one-prey kill with **all nearby prey**, and removed its 10 kg grass cap. Fixed source-path calculation, PowerShell execution-policy invocation, Java argument quoting, and UTF-16 log reading. An early GAMA null-clock/null-value error prompted serial execution (**parallel: false**); final tests passed. Boundary-cell choice and movement-cost assumptions were documented.

Final verification and outputs

Acceptance tests: lion energy 49.95 → 49.90 → 49.85 without prey; two co-located zebras killed, energy 69.95; starvation at energy 0.025; grazing never negative; grass rose beyond 10 kg. A 200-step extended run (seed 42) ended with **20 lions, 10 zebras, mean grass 24.061 kg/cell, mean lion energy 45**; animals stayed within the world. Final map/charts inspected; no runtime exceptions in final runs.

Deliverables & transparency

Generated **SPEC.md, savannah_baseline.gaml, savannah_extended.gaml, COMPARE.md, LINE_BY_LINE.md, REPORT.pdf** and this log; **COMMITS.txt** documents Git history. The one-page PDF report used a rasterized page layout. These tests demonstrate implementation against the specified rules, not ecological calibration or multi-seed generality. Sources consulted: existing Unit 1/specification, installed GAMA 2025.6.4 tools/logs, official GAMA Statements and Runtime Concepts documentation.